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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » The Joy of Computing using Python (course)

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Course outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1 ()

Week 2 ()

Week 3 ()

week 4 ()

Week 5 ()

Week 5: Assignment 5

Assignment not submitted

Due date: 2026-02-25, 23:59 IST.

Dilrez Builds a Smart Student Record System

Dilrez is a teaching assistant who is asked to create a **Smart Student Record System** for a small institute.

The system must store, update, analyze, and report student data efficiently.

To achieve this, Dilrez chooses **Python Dictionaries** because:

- Each student has a **unique roll number**
- Each roll number maps to **multiple details**
- Data needs to be **updated, searched, and processed quickly**

What Is a Dictionary? (Key–Value Mapping)

A dictionary stores data in **key : value** form.

```
student = {  
    "roll": 101,  
    "name": "Ayaan",  
    "age": 20  
}
```

2. Accessing Dictionary Data

```
print(student["name"])  
print(student.get("age"))
```

.get() is safer because it does not cause an error if the key is missing.

3. Adding & Updating Data in a Dictionary

- Introduction to Dictionaries (unit? unit=104&lesson=105)
- Speech to Text : No need to write 01 (unit? unit=104&lesson=106)
- Speech to Text : No need to write 02 (unit? unit=104&lesson=107)
- Speech to Text : No need to write 03 (unit? unit=104&lesson=108)
- Monte Hall : 3 doors and a twist 01 (unit? unit=104&lesson=109)
- Monte Hall : 3 doors and a twist 02 (unit? unit=104&lesson=110)
- Rock, Paper and Scissor : Cheating not allowed !! 01 (unit? unit=104&lesson=111)
- Rock, Paper and Scissor : Cheating not allowed !! 02 (unit? unit=104&lesson=112)
- Rock, Paper and Scissor : Cheating not allowed !! 03 (unit? unit=104&lesson=113)

Dilrez now wants to add marks.

```
student["marks"] = 85
student["age"] = 21
```

4. Dictionary of Lists (One Student, Multiple Scores)

Each student has marks in multiple tests.

```
student = {
    "roll": 101,
    "name": "Ayaan",
    "marks": [78, 85, 90]
}
```

Processing the list inside a dictionary

```
average = sum(student["marks"]) / len(student["marks"])
print(average)
```

5. Dictionary of Dictionaries (Structured Records)

Now Dilrez stores **multiple students**.

```
students = {
    101: {"name": "Ayaan", "marks": [78, 85, 90]},
    102: {"name": "Riya", "marks": [88, 92, 79]},
    103: {"name": "Kabin", "marks": [70, 75, 80]}
}
```

Accessing nested data

```
print(students[102]["name"])
print(students[101]["marks"][1])
```

This structure is ideal for **databases, JSON, APIs**.

6. List of Dictionaries (Row-Wise Data)

Sometimes data comes row-by-row (like Excel).



- ☐ Rock, Paper and Scissor : Cheating not allowed !! 04 (unit? unit=104&less on=114)
- ☐ Sorting and Searching : 20 questions game 01 (unit? unit=104&less on=115)
- ☐ Sorting and Searching : 20 questions game 02 (unit? unit=104&less on=116)
- ☐ Sorting and Searching : 20 questions game 03 (unit? unit=104&less on=117)
- ☐ Sorting and Searching : 20 questions game 04 (unit? unit=104&less on=118)
- ☐ Sorting and Searching : 20 questions game 05 (unit? unit=104&less on=119)
- ☐ Sorting and Searching : 20 questions game 06 (unit? unit=104&less on=120)
- ☐ Sorting and Searching : 20 questions game 07 (unit? unit=104&less on=121)
- ☐ Sorting and Searching : 20

```
students = [
    {"roll": 101, "name": "Ayaan", "marks": 85},
    {"roll": 102, "name": "Riya", "marks": 90},
    {"roll": 103, "name": "Kabin", "marks": 78}
]
```

Searching in list of dictionaries

```
for student in students:
    if student["roll"] == 102:
        print(student["name"])
```

7. Looping Through Dictionaries

Keys only

```
for key in student:
    print(key)
```

Values only

```
for value in student.values():
    print(value)
```

Key-Value pairs

```
for key, value in student.items():
    print(key, value)
```

Complete Mini Program



questions
game 08 (unit?
unit=104&less
on=122)

☐ Week 5
Feedback
Form: The Joy
of Computing
using Python
(unit?
unit=104&less
on=123)

☐ Week 5 :
Programming
Assignment 1
(/noc26_cs84/
progassignme
nt?name=623)

☐ Week 5 :
Programming
Assignment 2
(/noc26_cs84/
progassignme
nt?name=624)

☐ Week 5 :
Programming
Assignment 3
(/noc26_cs84/
progassignme
nt?name=625)

☐ Quiz: Week 5:
Assignment 5
(assessment?
name=632)

Week 6 ()

Text
Transcripts
()

Download
Videos ()

```
students = {
    101: {"name": "Ayaan", "marks": [78, 85, 90]},
    102: {"name": "Riya", "marks": [88, 92, 79]}
}

def class_average(students_dict):
    total = 0
    count = 0
    for student in students_dict.values():
        total += sum(student["marks"])
        count += len(student["marks"])
    return total / count

print(class_average(students))
```

1) Dilrez wants to extract the first test score of Riya from a nested student record. What will be printed? **1 point**

```
students = {
    101: {"name": "Ayaan", "marks": [78, 85, 90]},
    102: {"name": "Riya", "marks": [88, 92, 79]}
}

print(students[102]["marks"][0])
```

- ☐ 92
☐ 79
☐ 88
☐ Error

2) A new test score is added for Kabir. How many marks does Kabir finally have? **1 point**

```
student = {
    "name": "Kabir",
    "marks": [70, 75, 80]
}

student["marks"].append(85)
print(len(student["marks"]))
```

- ☐ 3
☐ 85
☐ 4
☐ Error

3) Dilrez modifies a list stored inside a dictionary and then reuses the same reference. What is printed? **1 point**



```
data = {"nums": [1, 2, 3]}
ref = data["nums"]

ref.append(4)
print(data["nums"])
```

- ☐ [1, 2, 3]
- ☐ [1, 2, 3, 4]
- ☐ Error
- ☐ [4]

4) Dilrez slices a list obtained from a dictionary and modifies the slice. What happens to the original list? **1 point**

```
student = {"marks": [10, 20, 30, 40]}
subset = student["marks"][:2]
subset.append(99)

print(student["marks"])
```

- ☐ [10, 20, 30, 40]
- ☐ [10, 20, 99]
- ☐ Error
- ☐ [10, 20]

5) A function reassigns a dictionary parameter instead of mutating it. What value is printed? **1 point**

```
def update(d):
    d = {"x": 100}

data = {"x": 10}
update(data)
print(data["x"])
```

- ☐ 10
- ☐ 100
- ☐ None
- ☐ Error

6) Dilrez modifies a dictionary while iterating over its keys *without creating a copy*. What happens when the code runs? **1 point**



```
data = {"a": 1, "b": 2}

for k in data.keys():
    if k == "a":
        data["c"] = 3

print(len(data))
```

- ☐ 2
- ☐ 3
- ☐ 1
- ☐ Runtime Error

7) Dilrez draws a bar graph to compare students fairly, even though they have attempted a different number of tests. Which student's bar will be the tallest?

1 point

```
students = {
    "Ayaan": [70, 80, 90],
    "Riya": [85, 90],
    "Kabir": [60, 65, 70, 75]
}
```

- ☐ Kabir
- ☐ Ayaan
- ☐ Riya
- ☐ All bars will be equal

8) Dilrez plots a line graph showing performance of each test across all students. Which test appears as the highest point?

1 point

```
students = {
    1: {"marks": [50, 60, 70]},
    2: {"marks": [60, 70, 80]},
    3: {"marks": [70, 80, 90]}
}
```

- ☐ Test 2
- ☐ Test 1
- ☐ Test 3
- ☐ All tests overlap

9) Dilrez creates a pie chart showing pass vs fail distribution (pass ≥ 60). Which section occupies more area?

1 point



```
records = [  
    {"name": "A", "marks": 45},  
    {"name": "B", "marks": 75},  
    {"name": "C", "marks": 65},  
    {"name": "D", "marks": 55},  
    {"name": "E", "marks": 85}  
]
```

- ☐ Equal distribution
- ☐ Pass section
- ☐ Cannot be inferred
- ☐ Fail section

10) Dilrez plots a cumulative trend graph of weekly progress. What overall shape will the graph show? **1 point**

```
data = {  
    "week1": [10, 20],  
    "week2": [30],  
    "week3": [40, 50]  
}
```

- ☐ Decreasing trend
- ☐ Flat line
- ☐ Random fluctuations
- ☐ Increasing trend

Searching for a Book ID in a Digital Library.

A college library maintains a **sorted list of book IDs**.

Each time a student requests a book, the system must **search for the book ID** and also **measure how many checks (iterations)** were needed to find it.

Two interns, **Aarav and Neel**, suggest two different approaches:

- Aarav uses **Linear Search**
- Neel uses **Binary Search**

The librarian wants to understand:

1. How each method works
2. How many iterations each method takes
3. Which method is more efficient and *why*

Part 1: Linear Search Case Study

Idea Behind Linear Search



Linear search works exactly like **checking books one by one on a shelf**, starting from the first book until the required one is found.

- No assumptions about sorting
- Simple and intuitive
- Works for **any list**

```
def linear_search(a, x):  
    count = 0  
  
    for i in range(len(a)):  
        count += 1  
        if a[i] == x:  
            print(f"Element found at position {i}.")  
            return count  
  
    print("Number is not present.")  
    return count
```

Sample Scenario

a = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

x = 7

We are searching for the element **7** in the list:

The algorithm starts from the **first index (0)** and moves forward **one element at a time**.

1. First iteration

The element at index 0 is 1.

It does not match 7, so the counter increases and the search continues.

2. Second iteration

The element at index 1 is 2.

Still not a match. The algorithm moves ahead.

3. Third iteration

The element at index 2 is 3.

Again, no match.

4. Fourth iteration

The element at index 3 is 4.

The search continues.

5. Fifth iteration

The element at index 4 is 5.

No match yet. 6.

6. Sixth iteration

The element at index 5 is 6.

The algorithm proceeds further.



7. Seventh iteration

The element at index 6 is 7.

This time, the element matches the target.

Why the iteration count is 7

The counter increases **before checking each element**, and the algorithm must **visit every position from index 0 to index 6**.

So, even though the element is at position 6, it takes **7 checks** to find it.

Part 2: Binary Search Case Study

Idea Behind Binary Search

Binary search is like **opening a dictionary**:

- You don't start from page 1
- You open the **middle**, then decide left or right

Important Requirement:

The list **must be sorted**

```
def binary_search(a, x):
    first_pos = 0
    last_pos = len(a) - 1
    flag = 0
    count = 0

    while first_pos <= last_pos and flag == 0:
        count += 1
        mid = (first_pos + last_pos) // 2

        if x == a[mid]:
            flag = 1
            print(f"Element found at position {mid}.")
            return count
        elif x < a[mid]:
            last_pos = mid - 1
        else:
            first_pos = mid + 1

    print("Number is not present.")
    return count
```

Same Scenario, Different Strategy

a = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

x = 7

Binary search works by **eliminating half of the list** after every comparison.

1. First iteration

The middle element of the list is calculated.

The value at the middle position is 5.

Since 7 is greater than 5, the algorithm **ignores the left half** and continues with the right half.

2. Second iteration

A new middle is calculated in the remaining right portion.

The value now checked is 8.

Since 7 is smaller than 8, the algorithm **moves left**.



3. Third iteration

The middle is recalculated again.

The value checked is 6.

Since 7 is greater than 6, the algorithm shifts right.

4. Fourth iteration

The middle position now contains 7.

The target is found, and the search stops.

Why the iteration count is only 4

Each step removes **half of the remaining elements**, drastically reducing the number of comparisons.

This is why binary search grows logarithmically and is much faster for large datasets.

11) A list contains 10 sorted elements. The target element is located at index 6. **1 point**
Using the **linear search code from the case study**, how many iterations will be counted?

- ☐ 10
- ☐ 6
- ☐ 7
- ☐ 5

12) In the binary search implementation, what primarily causes the number of iterations **1 point** to reduce compared to linear search?

- ☐ The use of a flag variable
- ☐ The loop runs fewer times due to index jumps
- ☐ The list length is fixed
- ☐ Elements are compared sequentially

13) If the searched element is **not present** in the list of 10 elements, which statement is **1 point** true regarding the iteration count?

- ☐ Linear search will stop after half the list
- ☐ Binary search will always run exactly 10 times
- ☐ Binary search cannot handle missing elements
- ☐ Linear search will check all elements

14) If the number of elements increases from 10 to 1,000, which graph best represents **1 point** the growth of iterations for **binary search**?

- ☐ A slowly rising curve
- ☐ A flat horizontal line
- ☐ A zig-zag pattern
- ☐ A steep straight line

15) In the binary search code, what would happen if the list were **unsorted** but the same logic was applied?



- ☐ The algorithm would still work but slower
- ☐ The iteration count would double
- ☐ Python would raise a runtime error
- ☐ The element may not be found even if present

You may submit any number of times before the due date. The final submission will be considered for grading.

Submit Answers

